Ayush Srivastava

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Education

University of San Francisco — **San Francisco, CA**
M.S. in Data Science — *July 2025 - July 2026*

Birla Institute of Technology and Science, Pilani Campus — **Pilani, India**
M.Sc. Mathematics & B.E. Engineering — *Aug 2016 – May 2021*

Technical Skills

Tools / Packages: C/C++, Python, PyTorch, Hugging Face Transformers, QLoRA, BitsAndBytes, FAISS, Hugging Face, Cosine Similarity, AWS, Redshift, Matillion, SQL, MATLAB, Linux, Tableau, Scikit-Learn, Numpy, Pandas, Matplotlib, OS, NLTK, Dlib

Statistical Models: Linear / Logistic Regression, Decision Trees, Random Forests, K-means, Principal Component Analysis.

Experience

Data Scientist Intern — **San Francisco Bay Area, CA (Remote)**
LexisNexis USA — *Dec 2025 – Present*

- **Deployed LLM**-powered summarization system via **Python** and **proxy API**, raising performance score: **95.74%** to **100%**
- **Benchmarked** Claude Sonnet vs Haiku on cost, output quality, latency on **20+** test cases, increasing accuracy: **75% to 91%**
- **Built** batch script for **100+** PDF, DOCX, HTML files via **pdfplumber**, **docx2txt**, **BeautifulSoup**, cutting prep by **90%**.
- Extended production pipeline with a new **Python** task in the processing chain, generating and mapping UUIDs to route dual **LLM** calls and deliver structured outputs to frontend.
- Drove accuracy improvements by incorporting feedback cycles with SMEs to iteratively refine **LLM** outputs across **2+** projects, .

Senior Data Engineer — **Gurgaon, India**
Axtria-Ingenious Insights — *May 2023 – May 2025*

- **Led ETL development and feature engineering** across **6+ domains**, **productizing API wrappers** within an in-house ETL product to integrate **10+ reports** into Tableau, ensuring scalable, high-quality data architecture.
- **Processed and modeled 20M+ records** from **8+ vendors**, applying **statistical profiling and data mining techniques** to generate **30+ actionable insights** that informed strategy for **6+ pharmaceutical products**.
- Optimized data ingestion pipelines, boosting efficiency by **40%** via automation with **SQL**,**Python**,**Matillion**,**AWS Redshift**
- Collaborated with cross-functional teams to automate and integrate the delivery of **30+** Excel-based reports into **5+** dynamic, real-time dashboards using **Python** and **SQL**, ensuring seamless data flow and reporting automation.
- Performed data quality validation with **100+** test cases in **JIRA** to ensure defect-free pipelines.

Data Engineer — **Gurgaon, India**
Axtria-Ingenious Insights — *Jul 2021 – Apr 2023*

- **Spearheaded a team of 3 developers** to build Tableau dashboard visualizing **20+ KPIs** for **4+** teams improving sprint planning efficiency by **60%**; **Led 40+ client discussions** from a team of **50+** in 4 projects aligning business requirements
- **Enhanced BI data layers** by automating legacy reports, applying **data reconciliation**, improving accuracy by **15%** n

Data Analyst — **Gurgaon, India**
Axtria-Ingenious Insights — *Jul 2021 – Apr 2022*

- Independently handled weekly refresh of **100+ Key Performance Indicators (KPIs)** on **27+** dashboards embedding statistical aggregation methods to ensure accuracy for **35+ sales representatives**.
- Led 5+ weekly ad-hoc requests as per client needs using **exploratory data analysis (EDA)**, partnering with commercial excellence, incentive compensation, onshore and data quality teams.

Data Scientist Intern — **Remote**
Sequretek IT Solutions Pvt. Ltd. — *May 2020 – Jul 2020*

- Developed **deep learning** models (VAE, LSTM) for **unsupervised time-series anomaly detection** across **150+** predictors and **1M+** Active Directory (AD) event logs, leveraging **statistical profiling** and **temporal trend analysis** in **Python** to identify cyberattacks, achieving **KL-Divergence loss** of **2.23/4.60** (train/validation).

Teaching Assistant — **New Delhi, India**
Coding Ninjas Pvt. Ltd. — *July 2019 – November 2019*

- **Delivered 240+ hours** of instructional support to **50+ students** over **4 months** as a Teaching Assistant, explaining core **Data Structures and Algorithms** concepts in **C++** through live doubt-clearing and concept-teaching sessions.
- **Resolved 500+ student queries** across the complete Data Structures and Algorithms curriculum (arrays, stacks, queues, linked lists, trees, graphs, sorting, dynamic programming) through structured, step-by-step explanations.
- **Conducted** daily **3-hour sessions** combining one-on-one doubt resolution with group explanations, ensuring consistent coverage of weak areas across the batch.

Publications and Projects

ER Vital Signs Prediction | *LightGBM, GRU, PyTorch, Python* — Spring 2026

- Finished **2nd** in class competition forecasting ER patient vital signs on **14,420** observations across **1,596** patients; final submission used **30 LightGBM** models with zero tuned post-processing parameters.
- Engineered **520+** features with fold-stable selection and cross-observation aggregates; validated leakage via adversarial **AUC 0.9998** under **GroupKFold** CV by patient.
- Built two-model residual architecture with a soft-gate under **Huber loss**, combining a **GRU** sequence model and **LightGBM** ensemble; ran systematic ablations to lock final design.

Cook Bot Cuisine Classifier | *FastAPI, Docker, MLflow, GCP, Scikit-Learn* — Fall 2025

- Served cuisine classifier over **39,774** recipes and **20** cuisines via **FastAPI** and **Docker**, reaching **78.72%** accuracy; registered in **MLflow** on **GCP** and published to **Docker Hub**.

Scalable Fraud Detection Pipeline on GCP with Spark and Airflow — Jan 2026 - Mar 2026

- Trained and evaluated **4 Spark MLlib** classifiers (Logistic Regression, Decision Tree, Random Forest, GBT) on **6M+** transactions; **GBT** achieved **ROC-AUC 0.9994** on **1.27M** test rows.
- Engineered **8** balance-behavior features via **PySpark**, handling **0.13%** class imbalance through stratified undersampling; *orig_balance_diff* topped feature importance at **33%**.
- Built distributed training pipeline on **GCP Dataproc** with **Apache Spark**, orchestrated via **Airflow**, with raw data in **GCS** and predictions in **MongoDB Atlas**.

Legal Holding Generation using RAG Pipelines and Fine-Tuned LLMs — July 2025 - Aug 2025

- Developed a **RAG** pipeline on **LexGLUE** dataset using **Mistral 7B**, testing **3** retrieval strategies on **1,000** text input.
- Implemented **FAISS semantic retrieval** with **Legal-BERT**(**768 d**) embeddings, retrieving **3+** contexts per query in **RAG**.
- **Fine-tuned Mistral 7B** for generation on **LexGLUE** data (**1,000** records).**Cross-entropy loss** (Train: **1.46**, Test **1.63**).
- Quantized model to **4-bit** with **QLoRA**, training **33.5M** params, reducing size from **28GB** to **3.5GB** for efficient fine-tuning
- Fine-tuned LLM's cosine similarity: **0.8962**, RAG: Option 1:**0.8648**, Option 2:**0.8653**, Option 3:**0.8491**, on **200 test cases**.

Real Time Driver Drowsiness Detection Using GRU with CNN Features — Dec 2020

- Organized by: International Association of Pattern Recognition - Computer Vision and Image Processing 2020 — Springer, Singapore. DOI: https://doi.org/10.1007/978-981-16-1103-2_42
- Github: https://github.com/sayush2807/Driver-Drowsiness-Detection-using-GRU-with-CNN-features
- Built a **Custom Dlib Shape Predictor on iBUG-300W** to extract mouth region; **MAE** (Train**: 5.83,** Test: **11.83**).
- Trained 5 models: Inception-V3, DenseNet, Lenet, MobileNet, Feature Extractor on **AffectNet database (1M records)**.
- Developed Conv1D-GRU yawn detector achieving **99.99%** train and **99.97%** validation accuracy for real-time monitoring
- Achieved real-time performance of **30 FPS** on host and **23 FPS** on embedded board, ensuring in-vehicle system feasibility.
- Benchmarked **6+** facial datasets to select optimal sources for feature extraction, boosting model reliability and robustness.

Thesis | Image-Based Structural Damage Detection using CNN and Transfer Learning — Aug 2019 – Apr 2020

Supervisor: Prof. Ajit Pratap Singh

- **Developed** an image classification pipeline to detect **3+ structural damage types** using **CNNs** and **transfer learning** on a dataset of **2,500+ images** with an **80:10:10** train/validation/test split.
- **Compared 3 deep learning architectures**: a CNN (6 conv layers, 3 dense layers), VGG16 with 3 FC layers, VGG16 with **6 dense layers and 4 dropout layers**, reducing overfitting from **37%** gap to **8%** gap between train and test accuracy.
- **Applied data augmentation** to address severe class imbalance, increasing minority class images from **206 to 1,648** (flexural), **400 to 1,600** (shear), and **524 to 1,572** (combined), balancing distribution to **24-26%** per class.
- **Achieved 91.47% test accuracy** using optimized VGG16 with frozen convolutional weights and deeper FC layers, outperforming baseline custom CNN (**58.71%**) and shallow VGG16 (**60.66%**)
- **Trained** models using **TensorFlow/Keras** with **Adam optimizer** (learning rate **0.0001**, decayed by **0.1** every **10 epochs**), batch size **8**, over **30 epochs** on **NVIDIA GTX 1650** GPU, evaluated with confusion matrices across all **4 classes**.

Thesis | Traffic Flow Prediction using Deep Learning (LSTM, GRU, MLP) — Aug 2020 – May 2021

Supervisor: Prof. Ajit Pratap Singh

- **Developed** deep learning models for time series traffic flow prediction using **12,000+** sequential data points from California **PeMS** dataset, spanning **80+ days**, aggregated at **5-minute** intervals across a **70%/20%/10%** train/validation/test split.
- **Implemented and compared 3 architectures**: **MLP** (5 dense layers), **LSTM** (3 LSTM layers + 3 dense layers), and **GRU** (3 GRU layers + 3 dense layers) using **TensorFlow/Keras** with **NVIDIA GTX 1650** GPU acceleration.
- **Designed** supervised time series pipeline using **sliding window** technique (window size **8**) to transform **12,000+** sequential observations into input-output pairs, capturing temporal dependencies for next-step traffic flow prediction.
- **Achieved** best performance with **LSTM**: test RMSE **9.81**, R^2 **0.9398**, outperforming GRU (RMSE **10.08**, R^2 **0.9365**) and MLP (RMSE **11.03**, R^2 **0.9240**), reducing prediction error by **11%** over baseline MLP.

- **Trained** LSTM, GRU models for **50 epochs** (batch size **64**) vs MLP's **10 epochs** (batch size **512**), using **Adam optimizer** (learning rate **0.001**), **tanh** activation for recurrent layers and **ReLU** for dense layers, evaluated with **RMSE** and $\mathbf{R^2}$ metrics.

Comparative Analysis of Neural Networks and ANFIS in Water Storage Prediction — Jan 2020 – May 2020

- Collected weekly time series data of live storage of Mettur Dam, Chennai (2001-2013) to train 2 forecasting models.
- **Long Short-Term Memory (LSTM) model** using **Keras** in **Python**: Training: 2.44, Validation: 2.66, Testing: 7.87.
- **Adaptive Neuro-Fuzzy Inference System (ANFIS)** in MATLAB: Training: 4.42, Validation: 5.09, Testing: 8.12.

Face Generation, Udacity — September 2020 – October 2020

- Trained a **Deep Convolutional Generative Adversarial Network (DCGAN)** on the **CelebFaces Attributes Dataset (CelebA)** to generate realistic human faces.
- Implemented **Discriminator** and **Generator** architectures using **Convolution** and **Transpose Convolution** layers respectively in **PyTorch**.
- Achieved **Discriminator loss: 0.06** and **Generator loss: 0.56**, producing high-quality synthetic face images using the trained Generator.

Dog Breed Classification using CNNs and Transfer Learning (PyTorch) — Sep 2020 – Oct 2020

- Built an end-to-end **computer vision pipeline** using **Convolutional Neural Networks (CNNs)** and **transfer learning** in **PyTorch** to classify dog breeds from real-world images.
- Developed an intelligent image classification system capable of **detecting dog breeds and identifying resembling dog breeds for human faces**, demonstrating multi-class classification capability.
- Implemented deep learning models leveraging **pretrained architectures** and fine-tuned fully connected layers, improving classification performance and convergence speed.
- Integrated **Haar Cascade classifiers** for human face detection and combined with CNN-based dog breed classification to build a complete inference pipeline.
- Designed modular prediction pipeline deployable in **web/mobile applications**, enabling real-time inference on user-supplied images.

TV Script Generation using LSTM-based Recurrent Neural Networks — Aug 2020 – Sep 2020

- Developed a **text generation model** using **Recurrent Neural Networks (RNNs)** and **Long Short-Term Memory (LSTM)** architecture to generate realistic TV scripts from sequential text data.
- Trained deep learning models on the **Seinfeld TV scripts dataset (9 seasons)**, enabling the model to learn linguistic structure, dialogue patterns, and contextual dependencies.
- Implemented complete **NLP preprocessing pipeline**, including tokenization, vocabulary encoding, sequence batching, and supervised sequence prediction.
- Built and trained multi-layer **LSTM network** in **Python**, generating coherent multi-character dialogues and long-form script content.
- Designed inference pipeline to generate new scripts from learned probability distributions, demonstrating practical applications in **generative AI and language modeling**.

Certifications

Deep Learning Specialization, Deeplearning.ai, Coursera — Apr 2020 – May 2020

Machine Learning & DSA in C++, Coding Ninjas Pvt Ltd — May 2019 – Aug 2019

TensorFlow in Practice Specialization, DeepLearning.AI (Coursera) — May 2020

Position of Responsibility

Sports Secretary, Malviya Bhawan — Sports Council — BITS Pilani, India

August 2017 – May 2018

- Developed financial records for **10+ sports events**, ensuring **100% accuracy** and compliance with budgetary guidelines.
- Managed an annual budget of **2 lacs** to procure sports kits and equipment for **250+ hostel residents**, increasing participation by **30%**.
- Coordinated logistics and planned events for the **BITS Open Sports Meet**, attended by **500+ participants** from **10+ colleges**.
- Organized intra-hostel tournaments across **3 sports**, with participation from **80+ students** across **6 hostels**.

Achievements

1st Place, Cricket Tournament — **BITS Open Sports Meet (BOSM)**, BITS Pilani, 2018.

1st Place (2×), Badminton Tournament — **Intra-BITS Open Sports Meet (I-BOSM)**, 2017 and 2019.

Selected to represent **Mount Carmel School** at the **Kennedy Space Center, NASA**, Florida, USA, 2013.

Gold Medalist, **National Abacus Championship Tournament**, BrainoBrain, 2010.